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Introduction to Immunologically-Mediated Diseases

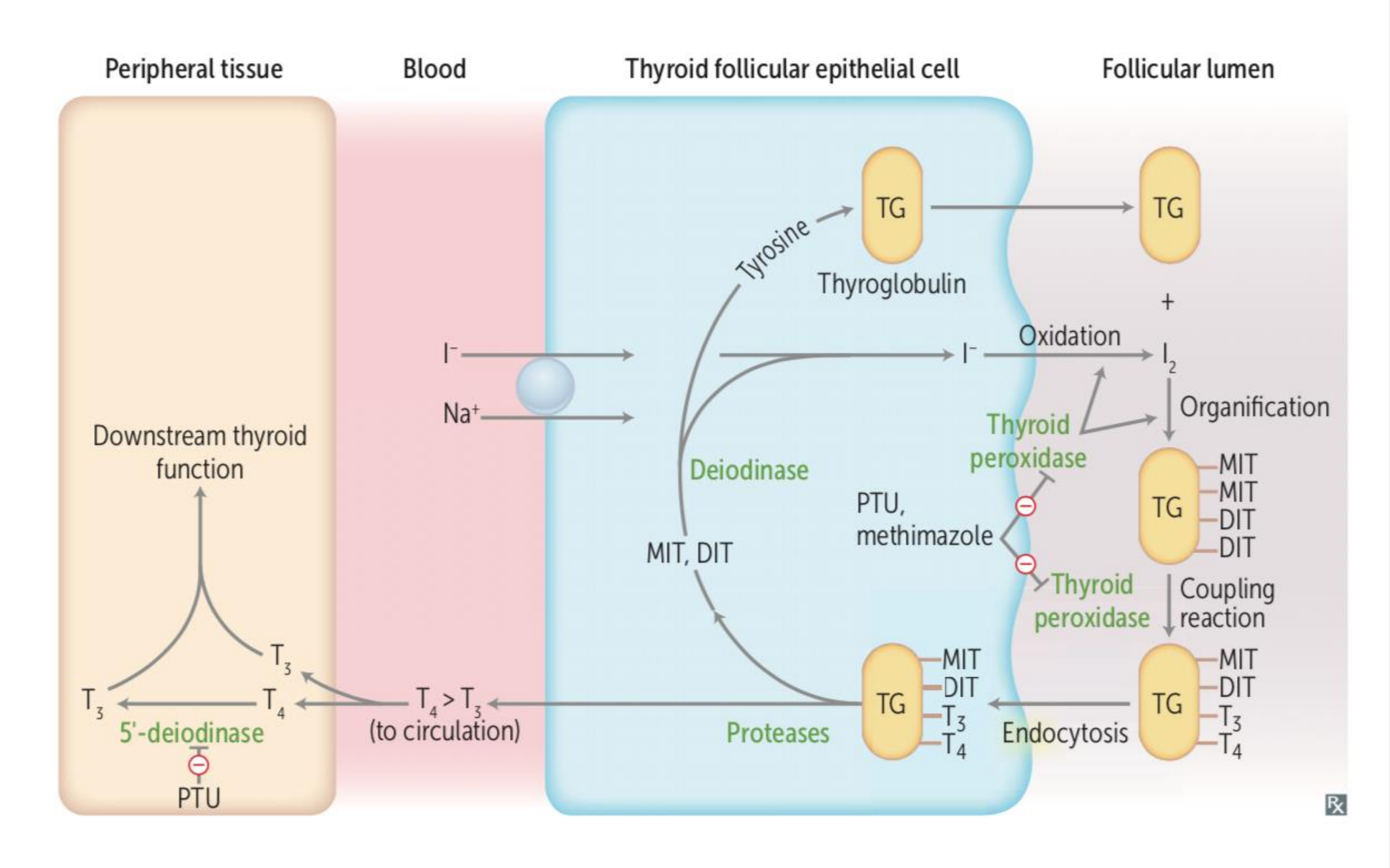
Shil Punatar, DO
Gastroenterology Fellow, PGY4
Franciscan Health, IL
shilpunatar@gmail.com

and

Tony A. Slieman, M.Sc., Ph.D.
Director of Academic Affairs and Associate Professor
Founding Director of Masters in Biomedical Sciences
Department of Biomedical and Anatomical Sciences
tslieman@nyit.edu

Autoimmune Diseases of Endocrine and Exocrine Glands

Hashimoto's Thyroiditis



Le, T., Bhushan, V., Sochat, M., & Chavda, Y. (2019). *First Aid for the USMLE Step 1 2019 A Student-to-Student Guide.* : McGraw Hill Education. 331.

Hashimoto's Thyroiditis

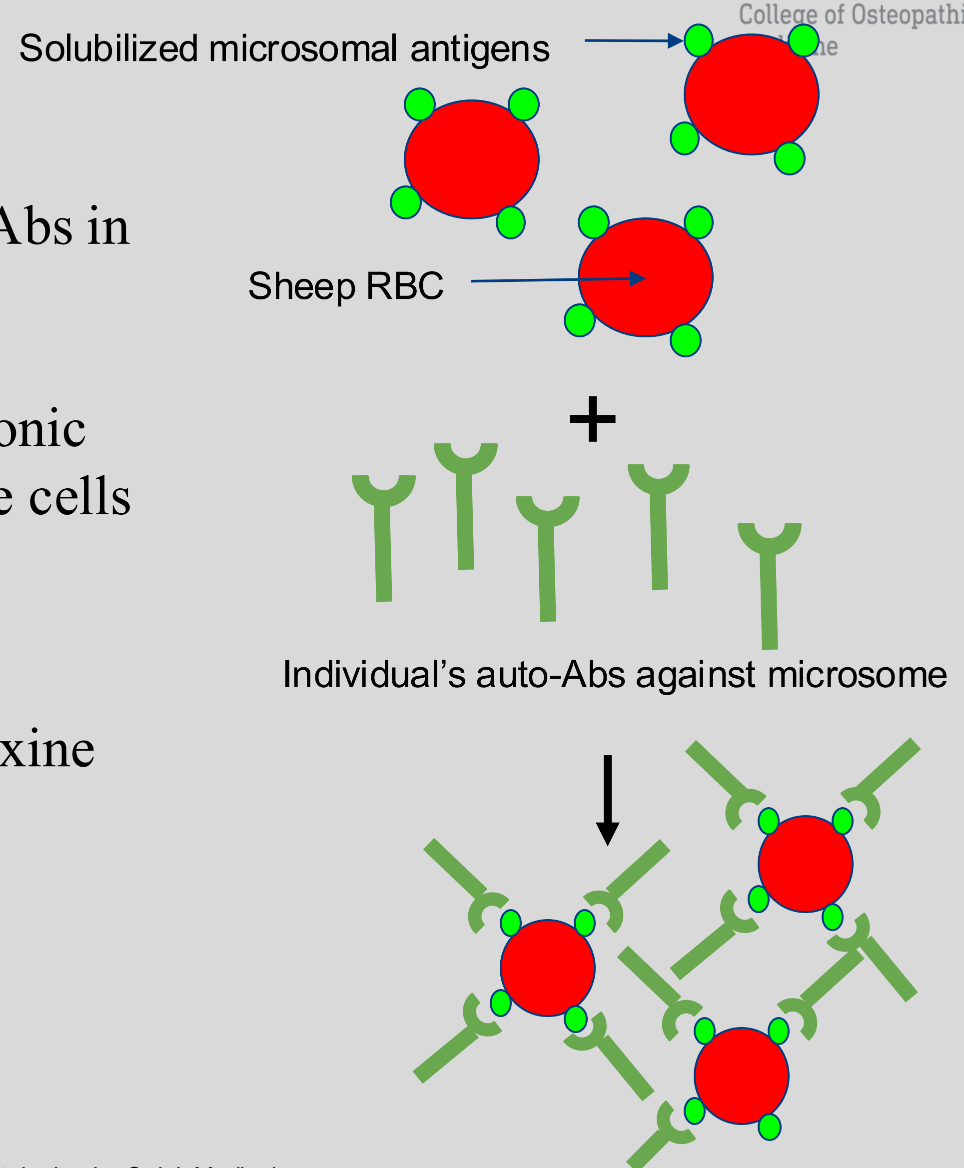
- Mechanism of Action: IgG auto-Abs against thyroid peroxidase (antimicrosomal) & thyroglobulin
- Organ-specific: Thyroid gland
- Signs & Symptoms:
  - Rick Factors: environmental (smoking, drugs), genetic (HLA-DR5 & DR3)
  - Diffuse, non-tender, moderate thyromegaly
  - Hypothyroidism

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Hashimoto's Thyroiditis

- <u>Diagnosis:</u>
  - Anti-TPO (microsomal) Abs in 90%(non-specific)
  - Anti-TG Abs in 40%
  - Follicle damage with chronic inflammation and Hurthle cells

- <u>Treatment:</u>
  - If hypothyroid, levothyroxine



Antimicrosomal Ab
hemagglutination assay

Thyroiditis. In: Papadakis MA, McPhee SJ, Bernstein J. eds. *Quick Medical Diagnosis & Treatment 2019*. New York, NY: McGraw-Hill; . http://accessmedicine.mhmedical.com/content.aspx?bookid=2566&sectionid=206896429. Accessed February 07, 2019.

Insulin-dependent Diabetes Mellitus

- Mechanism of Action: T-cell and Mϕ infiltration of β-cells
  - Decreased insulin
- Organ specific: Pancreatic β-cells
- Signs and Symptoms:
  - Genetic associations: HLA-DR3 & DR-4 Often asymptomatic
  - DKA common initial presentation
  - Polyuria, polydipsia, glucosuria

Jameson, J. L. (2016). *Endocrinology: Adult & pediatric*. Philadelphia, PA: Saunders/Elsevier.

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Insulin-dependent Diabetes Mellitus

- Diagnosis:
  - Oral blood glucose testing (fasting vs. non-fasting)
  - HbA1c ≥ 6.5% (monitoring)
  - Anti-glutamic acid decarboxylase auto-antibodies
  - Islet cell cytoplasmic antibodies
- Treatment:
  - Exogenous insulin
  - Pancreas transplant

Jameson, J. L. (2016). *Endocrinology: Adult & pediatric*. Philadelphia, PA: Saunders/Elsevier.

Sjögren's Syndrome

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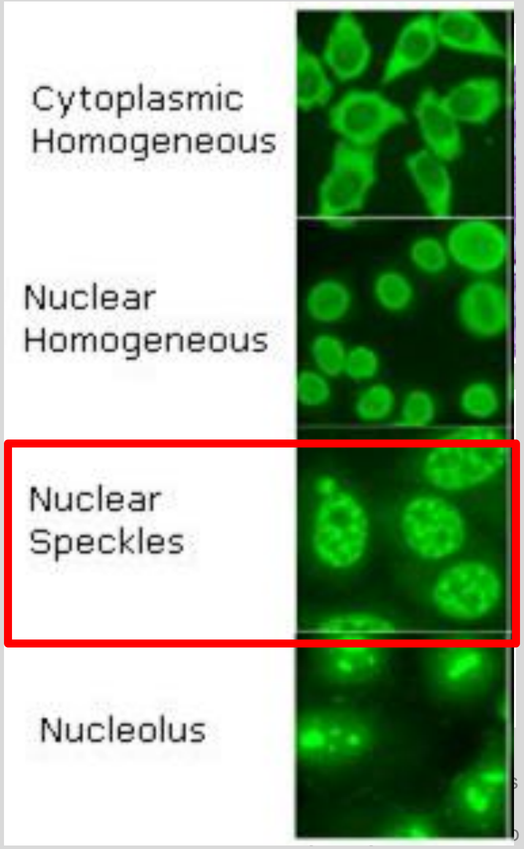
- Mechanism of Action: Lymphocytic infiltration + fibrosis → Destruction; 1º or 2º
- Non-organ specific: Exocrine glands, especially lacrimal & salivary
- Signs & Symptoms:
  - Xerostomia, keratoconjunctivitis sicca, b/l parotid enlargement
  - Vasculitis, xerosis, pruritus, & annular erythema
  - Marginal zone lymphoma
  - Rheumatoid Arthritis, Hashimoto's, Primary Biliary Cholangitis, Systemic Lupus Erythematosus

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Sjögren's Syndrome

- <u>Diagnosis:</u>
  - + RF 70%; + ANA: nuclear speckled; Anti-Ro/SSA & Anti-La/SSB
  - Schirmer test
  - Biopsy
- <u>Treatment:</u>
  - Immunosuppressive drugs (Prednisone)
  - Pilocarpine, artificial tears



WoBeM:
https://wwwn.cdc.gov/Nchs/Nhanes/2003-2004/SSANA_C.htm

Papadakis MA, McPhee SJ, Bernstein J. eds. *Quick Medical Diagnosis & Treatment 2019.* New York, NY: McGraw-Hill; . http://accessmedicine.mhmedical.com/content.aspx?bookid=2566&sectionid=206875738. Accessed February 17, 2019.

Case 1

A 15-year-old male presents to the ER with with tachypnea, labs demonstrate elevated glucose and a high anion gap metabolic acidosis. He reports no past medical history, though family reports ‘high sugars’ in several family members at a young age. What is the underlying pathological process?

A) Lymphocytic infiltration and fibrosis of exocrine glands

B) Cytotoxic T cells and Macrophages to β cells

C) Autoantibodies against IgG

D) Lymphocytic infiltration and fibrosis of endocrine glands

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Autoimmune Diseases of the Gastrointestinal Tract

Primary Biliary Cholangitis

- Mechanism of Action: etiology unclear; lymphocytic infiltrate + granulomatous inflammation
    - Auto-Abs against mitochondria
- Organ specific: medium-sized intrahepatic bile ducts
- Signs & Symptoms:
    - Pruritus
    - Obstructive jaundice
    - Cirrhosis, portal HTN
    - Xanthomas, steatorrhea

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

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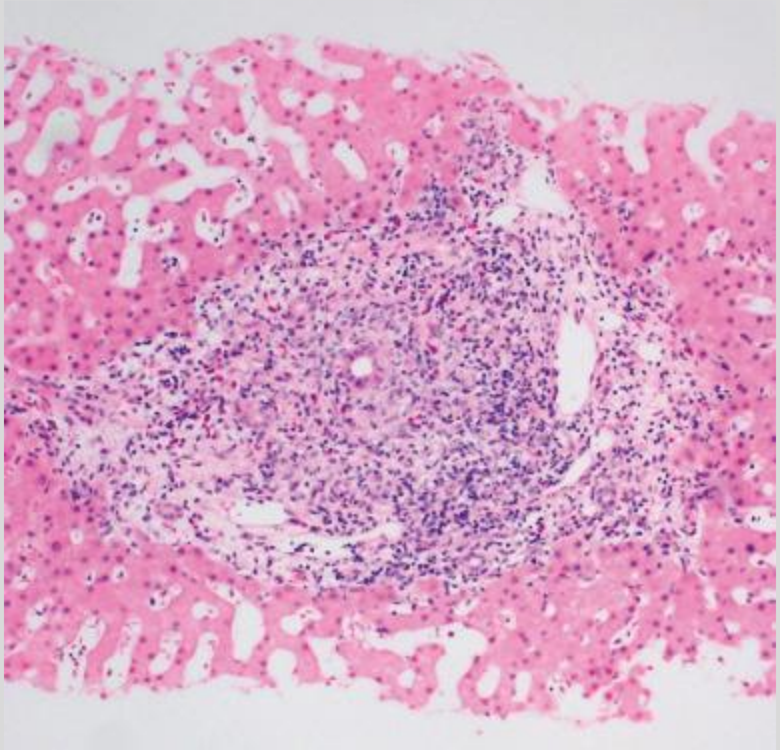
Primary Biliary Cholangitis

- <u>Diagnosis:</u>
  - Obstructive liver labs
    - ↑ Alk-Phos, GGT, LFT's, hypercholesterolemia; conjugated (direct) hyperbilirubinemia
  - +AMA 93%
  - Increased IgM
  - Liver biopsy
- <u>Treatment:</u> Ursodeoxycholic Acid (USDA)



A portal tract in primary biliary cirrhosis contains a nodular mixed inflammatory infiltrate consisting of lymphocytes, plasma cells, numerous eosinophils, and histiocytes.
Rosai and Ackerman's Surgical Pathology, 2018

Papadakis MA, McPhee SJ, Bernstein J. eds. *Quick Medical Diagnosis & Treatment 2019.* New York, NY: McGraw-Hill; . http://accessmedicine.mhmedical.com/content.aspx?bookid=2566&sectionid=206875738. Accessed February 15, 2019.

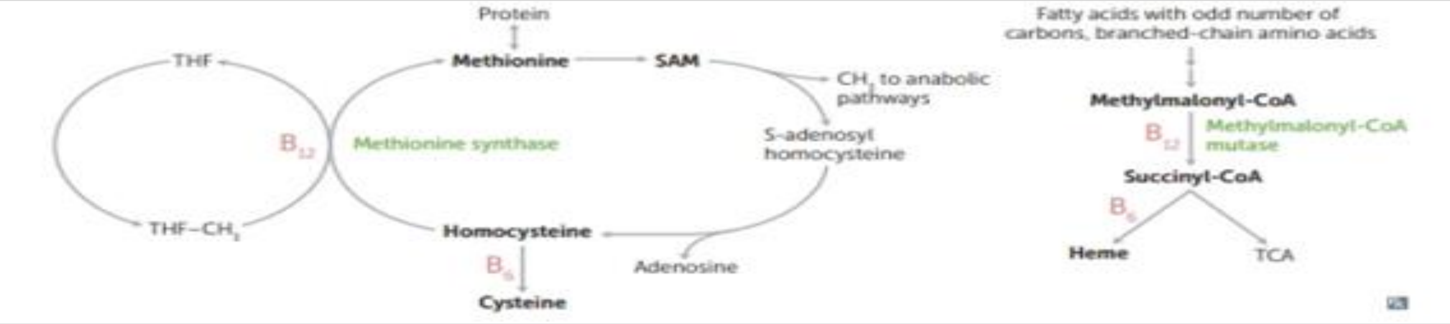
Pernicious Anemia

- Mechanism of Action: IgG auto-Antibodies against parietal cells &/or intrinsic factor
  - B12 (cobalamin) deficiency
- Organ specific: gastric parietal cells
- Signs & Symptoms:
  - Angular chelitis, glossitis
  - Peripheral sensory neuropathy
  - Loss of proprioception
  - Pallor, weakness, fatigue

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Pernicious Anemia

- Diagnosis:
  - Low B12, macrocytic (megaloblastic) anemia
  - Increased homocysteine and methylmalonic acid
  - Acid secretion analysis
  - Endoscopy with biopsy
  - Historically: Schilling test
- Treatment: Vitamin B12 **Intramuscular**



Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

First Aid for the USMLE Step 1 2019

Ulcerative Colitis

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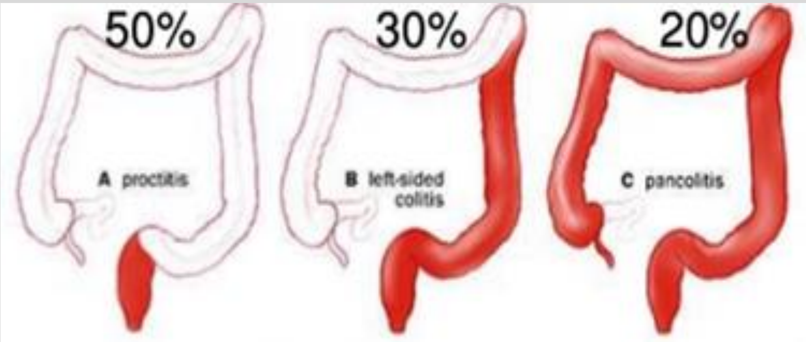
- Mechanism of Action: etiology unclear; mucosal & submucosal Th2-mediated inflammation
- Organ specific: Rectum



- Signs & Symptoms:
  - Smoking protective
  - Bloody diarrhea, Bright Red Blood Per Rectum (BRBPR), anemia, LLQ pain
  - Inflammation, oral ulcerations, pyoderma gangrenosum, 1º sclerosing cholangitis
  - Colorectal carcinoma

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier

https://doi.org/10.1016/j.crohns.2014.02.002

Ulcerative Colitis

Double contrast barium enema
https://radiopaedia.org/articles/lead-pipe-sign-colon?lang=us

- <u>Diagnosis:</u>
  - Inc ESR; dec Hgb/Hct & Albumin
  - +MPO/p-ANCA
  - Stool bacterial culture
  - Barium enema: Lead pipe sign
  - Endoscopic biopsy
- <u>Treatment:</u>
  - 5-Aminosalicylic acid preparations, sulfasalazine, 6-Mercaptopurine, infliximab
  - Colectomy

Kellerman, R. D., & Rakel, D. (2019). *Conns current therapy* 2019. Retrieved February 17, 2019, from https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323596480000559?scrollTo=#hl0000504

Case 2

A 22-year-old female presents to the clinic with concerns due to bloody diarrhea and what she describes as ‘cold sores’ at the corners of her lips that have not healed. She reports a recent camping trip but denies any fevers or chills. Labs are as follows: What is the potential disease process at hand?:

Lab tests	Patient’s Results	Normal Lab Values
Hemoglobin	10.0	12.0-18.0 g/dL
Hematocrit	31%	37-47%
Leukocyte count	8,000	4,500-11,000 mm
Erythrocyte Sedimentation Rate	50	0-20 mm in one hour
c-ANCA	-	-
p-ANCA	+	-

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Autoimmune Diseases of the Kidney

Poststreptococcal Glomerulonephritis

- <u>Mechanism of Action</u>: IgG, IgM, & C3 deposition on glomerular basement membrane (GBM) & mesangium
  - **Molecular mimicry:** Nephritogenic group A *Streptococci* pharyngitis or impetigo
- <u>Organ specific</u>: Renal GBM & mesangium
- <u>Signs & Symptoms:</u>
  - Ranges from asymptomatic to nephrotic syndrome
  - Children
  - Tea or Coca-cola colored urine
  - Periorbital & peripheral edema
  - HTN

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

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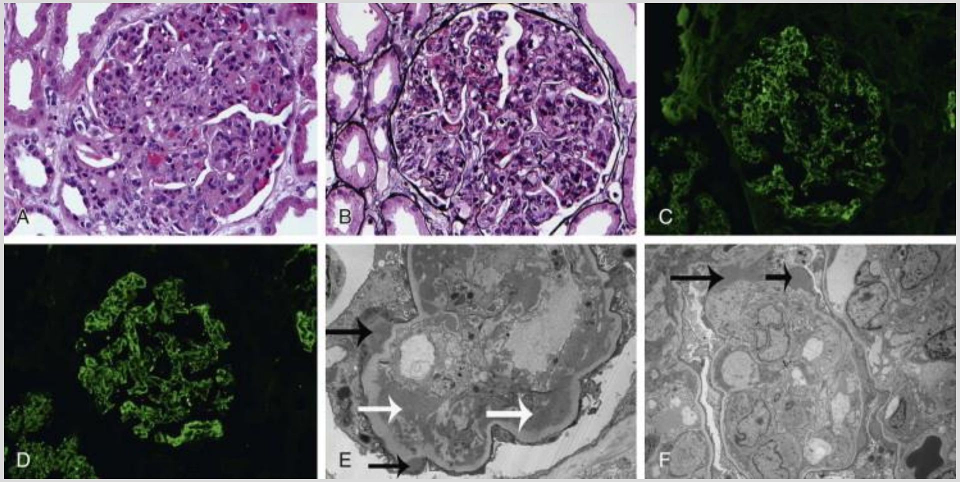
Poststreptococcal Glomerulonephritis

- <u>Diagnosis:</u>
  - Low C3 and/or CH50
  - ASO titers
  - Immunofluorescence microscopy: Granular
- <u>Treatment:</u>
  - Anti-hypertensives
  - Antibiotics
  - Diuretics/Na^{+} restriction
  - Supportive



Poststreptococcal Glomerulonephritis

Sethi et al., Glomerular Disease. Andreoli and Carpenter's Cecil Essentials of Medicine. Published January 1, 2016. Pages 314-328

McPherson, R. A., Pincus, M. R., & Henry, J. B. (2017). *Henrys Clinical diagnosis and management by laboratory methods*. Retrieved February 11, 2019, from https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323295680000541?scrollTo=#hl0001356

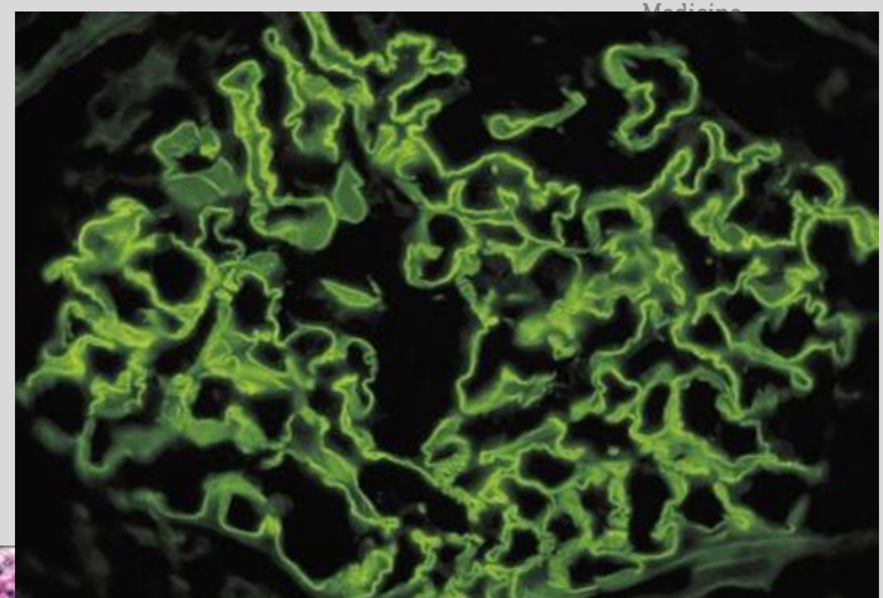
Goodpasture's Syndrome

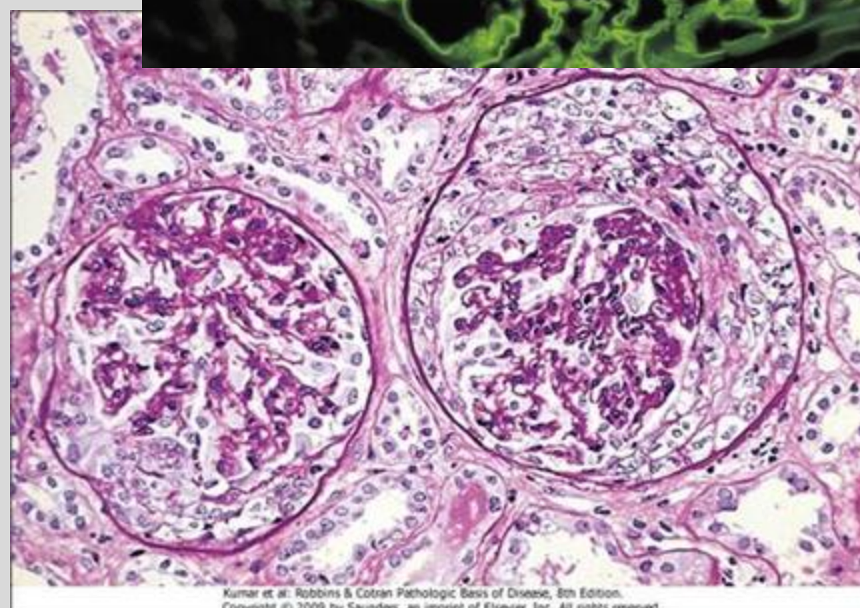
- Mechanism of Action: IgG auto-Abs against intrinsic Ag of GBM
- Organ specific: Renal GBM
- Signs & Symptoms:
  - Genetic associations: HLA-DR2
  - Adults
  - Rapidly progressive glomerulonephritis (hematuria)
  - Pulmonary hemorrhage (hemoptysis)
  - Preceding URI

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Goodpasture’s Syndrome

- **Diagnosis:**
  - Anti-GBM Abs >90%
  - CXR: pulmonary infiltrate
  - Light microscopy
  - Immunofluorescence microscopy: Linear
- **Treatment:**
  - Plasmapheresis
  - Cyclophosphamide
  - Corticosteroids





http://www.stepwards.com/?page_id=1190

https://microbeonline.com/elisa-test-for-antigenantibody-detection/

Autoimmune Diseases of the Skin

Pemphigus Vulgaris

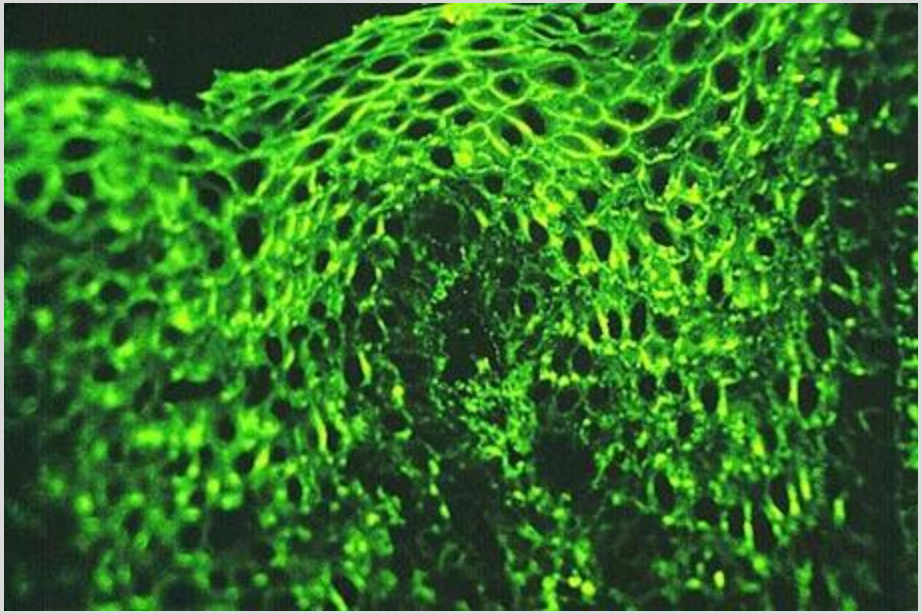
- Mechanism of Action: IgG auto-Abs against desmoglein 1 and/or 3 (desmosome)
- Organ-specific: Mucosal &/or cutaneous epithelial cell junctions
- Signs & Symptoms:
  - Initially, oral mucosa lesions
  - Painful, flaccid bullae
    - Positive Nikolsky sign

Lawrence L., James, W. D., et al: *Andrews' diseases of the skin*, ed 12, Philadelphia, 2016, *Elsevier.*

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Pemphigus Vulgaris

- <u>Diagnosis:</u>
  - Anti-desmoglein 1/3 Abs
  - Light microscopy: Acantholysis
  - Immunofluorescence: Reticular
- <u>Treatment:</u>
  - Corticosteroids
  - Rituximab



https://www.pathologystudent.com/whats-the-difference-between-pemphigus-vulgaris-and-bullous-pemphigoid/

Papadakis MA, McPhee SJ, Bernstein J. eds. *Quick Medical Diagnosis & Treatment 2019.* New York, NY: McGraw-Hill; . http://accessmedicine.mhmedical.com/content.aspx?bookid=2566&sectionid=206875738. Accessed February 12, 2019.

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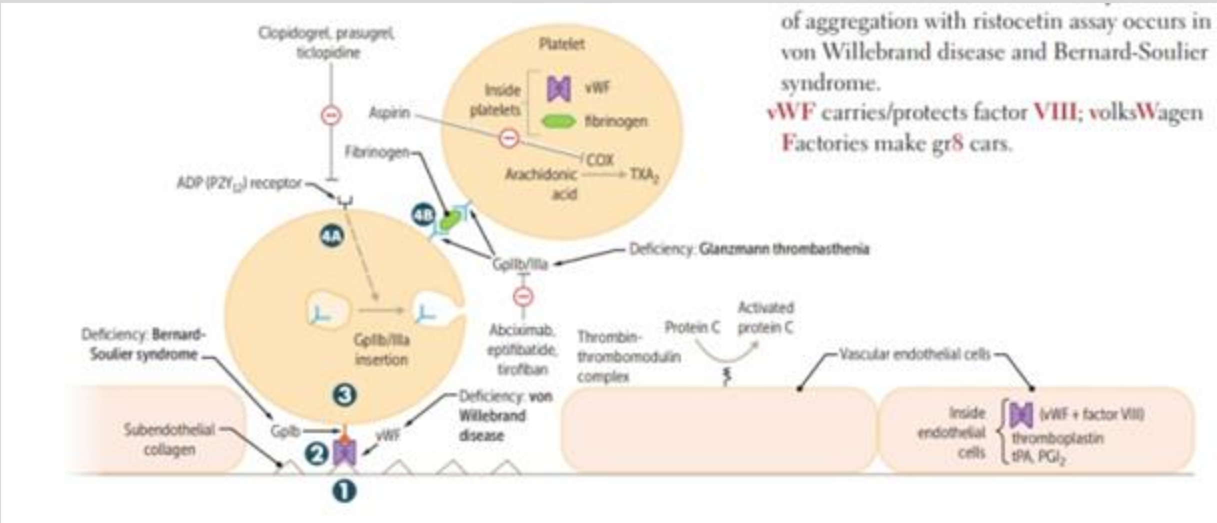
Autoimmune Diseases of the Vascular System

Immune Thrombocytopenic Purpura

- Mechanism of Action: Auto-Abs against GpIIb/IIIa
- Organ specific: Blood, platelets
- Signs & Symptoms:
  - Purpura, mucocutaneous bleeding, splenomegaly
  - Pallor, fatigue



First Aid for USMLE Step 1 2019

Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Immune Thrombocytopenic Purpura

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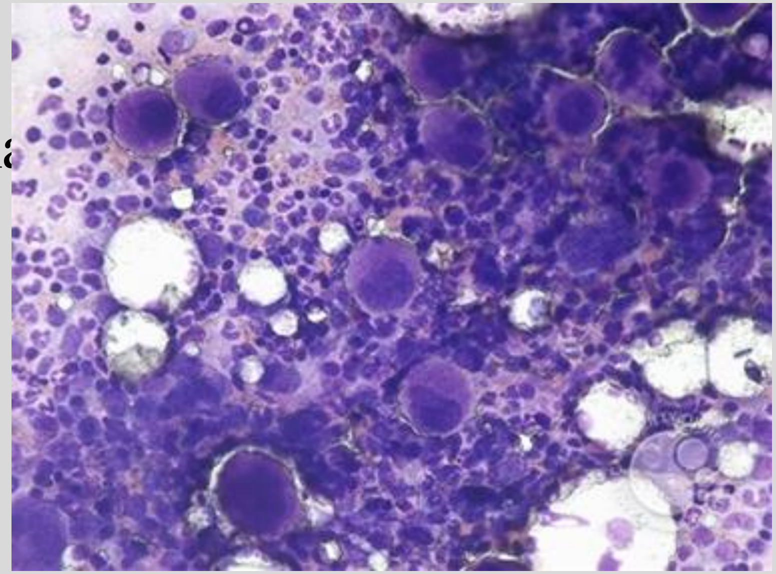
- <u>Diagnosis:</u>
  - CBC: Isolated thrombocytopenia
  - Diagnosis of exclusion
  - Blood smear
  - Bone marrow biopsy (adults)



https://imagebank.hematology.org/image/16992/bone-marrow-itp

Platelets below 50,000

- <u>Treatment:</u>
  - Children: none
  - Adults: Corticosteroids, rituximab, IVIG, & anti-IgD; platelets, splenectomy
  - D/C drugs inhibiting platelet function

Papadakis MA, McPhee SJ, Bernstein J. eds. *Quick Medical Diagnosis & Treatment 2019.* New York, NY: McGraw-Hill; . http://accessmedicine.mhmedical.com/content.aspx?bookid=2566&sectionid=206875738. Accessed February 14, 2019.

Antiphospholipid Syndrome

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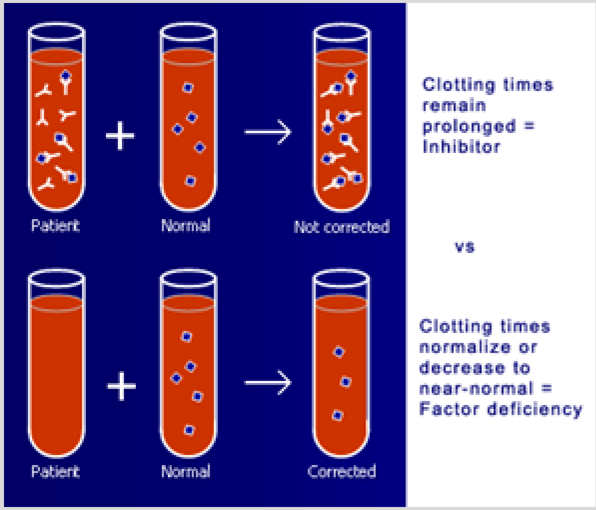
- <u>Mechanism of Action</u>: Auto-Abs vs. proteins that bind phospholipids → thrombin (acquired hypercoagulability; 1’ or 2”)
  - Lupus anticoagulant
  - Cardiolipin (IgG or IgM)
    - β_2-glycoprotein I (IgG or IgM)
- <u>Non-organ specific</u>: Hematologic; catastrophic APS ≥ 3 organ systems
- <u>Signs & Symptoms:</u>
  - Recurrent thrombosis: DVT, PE, CVA, MI, etc.
  - Spontaneous abortions
  - Livedo reticularis


https://healthjade.com/mottled-skin/

Hoffman, R. (2018). *Hematology: Basic Principles* (7th ed.). Retrieved February 20, 2019, from https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323357623001402?scrollTo=#hl0000431

Antiphospholipid Syndrome

- <u>Diagnosis:</u>
  - Sapporo APS Classification Criteria
    - Labs, ≥1: + Anticardiolipin (IgG and/or IgM), + Lupus Anticoagulant,+ β_2-glycoprotein I
    - Clinical ≥1: Vascular thrombosis, pregnancy morbidity



- <u>Treatment</u>: lifelong anticoagulation with warfarin (heparin if pregnant)

Hoffman, R. (2018). *Hematology: Basic Principles* (7th ed.). Retrieved February 20, 2019, from https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323357623001402?scrollTo=#hl0000431

Summary Slide

Diseases	AB	CLINICAL SIGNS
Hashimoto’s Thyroiditis		
T1DM		
Sjogren’s Syndrome		
PBC		
Pernicious Anemia		
UC		
PSGN		
Goodpasture’s Syndrome		
Pemphigus Vulgaris		
ITP		
Antiphospholipid Syndrome		

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